Md Sadekul Islam

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Research Interests

- **AI & Decision Intelligence:** Deep Reinforcement Learning, Graph Neural Networks, Learning-Augmented Combinatorial Optimization
- **Smart Manufacturing & Logistics:** Facility Layout Optimization, Digital Twins, Discrete-Event Simulation, Supply Chain Resilience
- **AI-Driven Engineering Design:** Additive Manufacturing, Design for Manufacturing, Simulation-Driven Product Design
- **Human-Centered Systems:** Human–Robot Collaboration, Safety in Collaborative Workcells, Ergonomic Design

Education

M.Engg. Advanced Engineering Management *Dhaka, Bangladesh*
Bangladesh University of Engineering and Technology (BUET) *Sept 2026*
- Thesis Title: Integrating Evolutionary Algorithm and Deep Reinforcement Learning for Efficient Facility Layout Design
- Advisor: Dr. Ridwan Al Aziz, Associate Professor, Dept. of Industrial and Production Engineering
- GPA: 3.45 / 4.00

B.Sc. Industrial and Production Engineering *Khulna, Bangladesh*
Khulna University of Engineering and Technology (KUET) *March 2019*
- Thesis Title: Improvement of Production Line Efficiency Using Tecnomatix Simulation Software and RPWM Line Balancing
- Advisor: Dr. Mahmud Parvez
- GPA: 3.59 / 4.00; ranked 11th of 54

Research Experience

Graduate Researcher, Learning-Based Facility Layout Design *Dhaka, Bangladesh*
Bangladesh University of Engineering and Technology *M.Engg. Thesis*
- Built a two-stage hybrid optimizer: a genetic algorithm generates feasible layouts, then a Rainbow DQN agent refines them
- Designed a 12-component composite reward, action masking and multi-cell overlap prevention; pre-trained a GNN encoder on synthetic material-flow graphs
- Validated on a 26-machine industrial layout, with sensitivity studies on reward weights, grid resolution and flow density
- Identified open problems (training instability, residual overlaps, sparse-flow reliability) that motivate constraint-aware learning methods
- Tools: Python, PyTorch, PyTorch Geometric, TensorFlow, Gymnasium, SimPy

Undergraduate Researcher, Simulation-Based Production Line Improvement *Khulna, Bangladesh*
Khulna University of Engineering and Technology *B.Sc. Thesis*
- Time-studied a bag-manufacturing sewing line, rebalanced it with the Ranked Positional Weight Method (RPWM), and validated the new balance in Siemens Tecnomatix Plant Simulation
- Published in 2019; extended with stochastic simulation, multi-criteria selection and lean improvement in a 2026 manuscript submitted to the *International Journal of Production Research*

Work Experience

Mechanical Product Designer, Refrigerator R&D *Gazipur, Bangladesh*
Walton Hi-Tech Industries PLC *2019 - Present*
- Led the design and July 2023 launch of Walton's largest side-by-side refrigerator (646 L, non-frost; 250+ parts)
- Designed the cabinet liner and the water-dispenser and ice-tray mechanisms: tight-tolerance assemblies combining electrical, plastic, potable-water and spring components
- Verified stability, reliability and performance through simulation and laboratory tests (push-pull, drop, snap-fit, steady-state thermal)
- Performed draft analysis on injection-molded HIPS and POM parts and ran mold trials, tuning mold temperature, injection parameters and gate sequencing
- Verified fit and function with 3D-printed prototypes before committing to mold design
- Benchmarked competitor designs, ran volume, weight and cost analyses, released toleranced 2D drawings and reviewed pilot-production samples
- Built Power BI dashboards to track R&D project and team KPIs; led the team that developed them

Industrial Trainee *Savar, Bangladesh*
Brothers Furniture Ltd. *2018*
- 3D modeling and design changes in AutoCAD and SolidWorks; 3D scanning for reverse engineering
- Monitored production quality; exposure to CNC and robotic production

Honors & Awards

Souvenir for Outstanding Performance, Walton Hi-Tech Industries PLC.

1st Runner-Up, Advance Engineering Management Challenge (AEMC), Khulna University of Engineering and Technology, February 2018.

Technical Board Merit Scholarship, Khulna University of Engineering and Technology, all eight semesters, 2015–2019.

Publications

Journal Publications

1. **Islam, M. S.**, Sarker, S., Parvez, M., "Production Efficiency Improvement by Using Tecnomatix Simulation Software and RPWM Line Balancing Technique: A Case Study," *American Journal of Industrial and Business Management*, 9(4), 809–820, 2019.

Journal Publications Under Review or in Final Preparation

1. **Islam, M. S.**, Sabuj, M. A., "Integrating Exact Line Balancing, Stochastic Simulation and Multi-Criteria Selection with Lean Improvement: Evidence from a Luggage Sewing Line." Status: Submitted to the *International Journal of Production Research*.
2. Sabuj, M. A.*, **Islam, M. S.***, Hridita, F. C., "Modeling Healthcare Supply Chain Risk Factors for Sustainable Development in an Emerging Economy: A Pythagorean Fuzzy DEMATEL Approach." Status: Submitted to *Expert Systems with Applications*. (*Co-first authors)
3. **Islam, M. S.**, Al Aziz, R., "Integrating Evolutionary Algorithm and Deep Reinforcement Learning for Efficient Facility Layout Design." Status: In final preparation.

Selected Design Projects

Prosthetic Leg
Fusion 360, KeyShot
- Lower-limb prosthesis with a spring-damped ankle module, designed for natural gait, comfort and cosmetic acceptability

Virtual Factory Design
SketchUp, V-Ray
- 3D facility models with equipment, storage, safety zones and operator positions to plan layout, collision avoidance and human–machine interaction

Portable Speaker
SolidWorks, Fusion 360, KeyShot, Autodesk Moldflow
- Screwless three-part snap-fit assembly designed for low weight and drop/vibration resistance; mold-flow analysis of the grille to guide material, process and tooling choices

Full CAD files and renders: sadik120.github.io

Technical Skills

CAD / CAE	SolidWorks (3D, motion, simulation), ANSYS, PTC Creo, Autodesk Fusion 360, AutoCAD, Autodesk Moldflow, KeyShot, SketchUp with V-Ray
Manufacturing	Injection-mold trials, draft/DFM analysis, 3D printing, toleranced 2D drawing release, 3D scanning for reverse engineering
Simulation & Optimization	Siemens Tecnomatix Plant Simulation, Arena, SimPy, Proplanner, LINGO, TORA, OR-Tools
Programming & ML	Python, C, C++, MATLAB; PyTorch, PyTorch Geometric, TensorFlow, Gymnasium, NumPy, Pandas
Data & Project Management	Power BI, Excel, Minitab, Oracle, Microsoft Project, Primavera P6, Git, LaTeX

English Proficiency

IELTS Academic (Nov 2023): Overall 7.0 — Listening 7.0, Reading 8.0, Writing 6.5, Speaking 6.0. Retake scheduled for November 2026.

References

Available upon request.